

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement 20: Healthcare & Telemedicine

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1. Problem Context

An emergency room bed management dashboard displaying real-time ward occupancy, prioritizing ICU transfers based on triage score, and orchestrating inter-facility ambulance transfer workflows.

Target Stakeholders / Actors: Triage Nurse, Hospital Administrator

2. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.	High	Pass: A patient with a higher acuity score automatically appears ahead of a lower-acuity patient. Fail: The system maintains FIFO order despite higher acuity.	Ensures critical patients receive priority for available beds.
FR-002	Functional	The system shall allow the Triage Nurse to update the status of each hospital bed as Occupied, Cleaning, or Available.	High	Pass: When a nurse changes a bed status, the new status is displayed on the dashboard. Fail: The previous status remains displayed.	Provides an accurate view of current bed availability.
FR-003	Functional	The system shall display real-time ward occupancy and available ICU beds to authorized users.	High	Pass: The dashboard displays current occupancy and available ICU beds after a bed-status update. Fail: The displayed information remains outdated.	Helps staff make timely bed allocation decisions.
FR-004	Functional	The system shall allow the Hospital Administrator to initiate and track inter-facility ambulance transfer requests for patients requiring transfer.	Medium	Pass: A transfer request can be created and its status changes as the transfer progresses. Fail: The request cannot be created or tracked.	Supports coordination of patients who need transfer to another facility.
FR-005	Functional	The system shall notify authorized hospital staff when an ICU bed becomes available for a high-priority patient on the waitlist.	High	Pass: An alert is generated when an ICU bed becomes available and a high-priority patient is waiting. Fail: No alert is generated.	Reduces delays in assigning critical patients to ICU beds.
NFR-001	Nonfunctional (Performance & Security)	Bed status transitions (Occupied, Cleaning, Available) must propagate to all connected ER dashboard clients in under 1 second via WebSockets.	High	Pass: Benchmark testing shows updates reach connected clients in less than 1 second under simulated peak load. Fail: Updates take 1 second or longer.	Ensures the dashboard provides timely information during emergencies.
NFR-002	Nonfunctional (Security)	The system shall restrict bed allocation, triage and transfer-management functions to authenticated and authorized users.	High	Pass: An unauthorized user attempting a restricted function is denied access. Fail: The restricted function is accessible without authorization.	Protects sensitive hospital operations and prevents unauthorized changes.

3. Actors and Use Cases

Actors

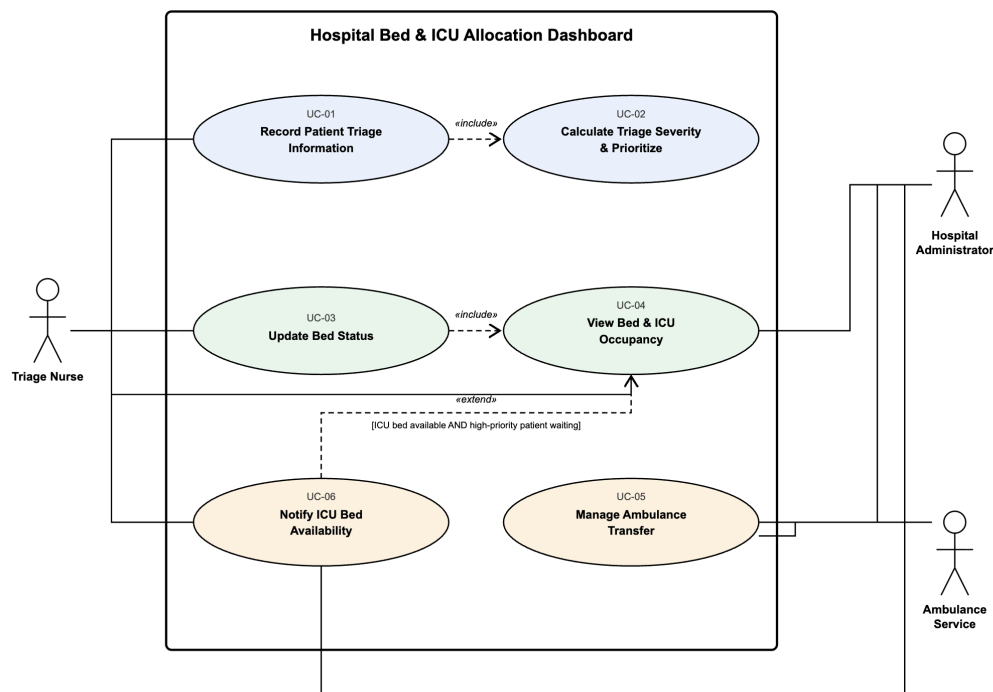
- 1. Triage Nurse:** Records patient triage information, updates bed status, and monitors bed/ICU allocation.
- 2. Hospital Administrator:** Monitors occupancy, manages inter-facility transfers, and receives ICU availability notifications.
- 3. Ambulance Service:** Receives and updates inter-facility ambulance transfer requests.

Use Cases

ID	Use Case	Primary Actor(s)
UC-01	Record Patient Triage Information	Triage Nurse
UC-02	Calculate Triage Severity & Prioritize	Triage Nurse
UC-03	Update Bed Status	Triage Nurse
UC-04	View Bed & ICU Occupancy	Triage Nurse, Hospital Administrator
UC-05	Manage Ambulance Transfer	Hospital Administrator, Ambulance Service
UC-06	Notify ICU Bed Availability	Triage Nurse, Hospital Administrator

4. UML Use-Case Diagram

The diagram below models all three actors, six use cases, associations, and both required « **include** » and « **extend** » relationships. The « **extend** » relationship is conditional: notification occurs when an ICU bed becomes available while a high-priority patient is waiting.



Associations:

Triage Nurse → UC-01, UC-03, UC-04, UC-06

Hospital Administrator → UC-04, UC-05, UC-06

Ambulance Service → UC-05.

« **include** » : UC-01 includes UC-02; UC-03 includes UC-04.

« **extend** » : UC-04 is extended by UC-06 when an ICU bed becomes available and a high-priority patient is waiting.

5. Use-Case Flow Specification

Use Case: UC-01 - Record Patient Triage Information

Goal: Record the patient's intake information, calculate the triage severity score, and place the patient at the correct position in the ER bed allocation waitlist.

Preconditions

The Triage Nurse is authenticated and authorized.

The patient has been registered in the hospital system.

The triage dashboard is available.

Postconditions

The patient's triage information is stored.

A triage severity score is calculated.

The patient is placed in the appropriate ER/ICU allocation priority order.

The updated priority information is available on the dashboard.

Main Success Scenario

1. Triage Nurse selects "Record Patient Triage Information".
2. System displays the patient intake form.
3. Triage Nurse enters the patient's relevant intake vitals and triage information.
4. System validates the entered information.
5. System calculates the patient's triage severity score.
6. System compares the patient's severity with the existing allocation queue.
7. System places the patient at the appropriate position in the bed allocation waitlist.
8. System displays the updated priority order to the Triage Nurse.
9. Use case ends successfully.

Alternate Flow A - Invalid Intake Information

- 4a. System detects missing or invalid intake information.
 - 4a1. System displays an error message identifying the invalid fields.
 - 4a2. Triage Nurse corrects the information.
 - 4a3. System validates the corrected information.

4a4. Flow resumes from Step 5 of the Main Success Scenario.

Alternate Flow B - Higher-Acuity Patient

7a. System calculates a higher severity score for the patient.

7a1. System moves the patient ahead of lower-acuity patients in the waitlist.

7a2. System updates the displayed priority order.

7a3. The use case continues successfully.

6. Summary

This Lab 1 identifies and specifies the functional and nonfunctional requirements for the Hospital Bed & ICU Allocation Dashboard and translates them into a UML use-case model and detailed use-case flow. The model focuses on real-time occupancy, triage-based prioritization, ICU availability notification, and inter-facility ambulance transfer coordination.